

0.1 Commutative Ring with Unity

Basic Preliminaries: Let R be a Commutative Ring with unity $1 \neq 0$, $A \subseteq R$ is subset, $I \subseteq R$ Ideal.

1. $(A) = RAR$
2. $I = R \iff I$ has a unit. (Not necessarily Commutative)
3. R is field $\iff$ only Ideal of R is 0 or R .
4. $P \subsetneq R$ is prime $\iff R/P$ is integral domain.
5. $M \subsetneq R$ is Maximal $\iff R/M$ is field.

Definition 0.1.1. Let R be an Integral domain. A map $N : R \rightarrow \mathbb{N} \cup \{0\}$ is called a **Norm** if $N(0) = 0$. If $N(a) > 0$ for all $a \neq 0$, then it is called a **positive Norm**.

Definition 0.1.2. Let R be a Commutative Ring, and $a, b \in R$ with $b \neq 0$.

1. a is called a **multiple** of b if there exists $x \in R$ s.t. $a = bx$. Simply, denote $b \mid a$.
2. A non-zero $d \in R$ is called a **greatest common divisor** of a and b if:

$$\begin{cases} d \mid a \text{ and } d \mid b \\ d' \mid a \text{ and } d' \mid b, \text{ then } d' \mid d \end{cases}$$

Since $b \mid a \iff a \in (b) \iff (a) \subseteq (b)$, d is a g.c.d. of a and b if:

$$(a, b) \subseteq (d) \text{ and } (a, b) \subseteq (d') \implies (d) \subseteq (d').$$

Proposition 0.1.3. $(a, b) = (d) \implies d$ is g.c.d. of a and b .

But the converse is false, In $\mathbb{Z}[x]$, 1 is g.c.d. of 2 and x but $R = (1) \neq (2, x)$.

0.2 Euclidean Domain (E.D.)

Definition 0.2.1. The integral domain R is said to be a **Euclidean domain** if there exists a Norm N s.t. for any $a, b \in R$ with $b \neq 0$, there exists $q, r \in R$ s.t. $a = qb + r$ where $N(r) < N(b)$ or $r = 0$.

Lemma 0.2.2. If R is an Euclidean domain with a Norm N , then for any $a, b \in R$ with $b \neq 0$, there exists a sequence $\{r_k\}_{k=0}^{\infty}$ satisfying: put $r_{-2} = a, r_{-1} = b$,

$$\begin{cases} \text{for } 0 \leq k \leq n+1, r_{k-2} = q_k \cdot r_{k-1} + r_k \text{ for some } q_k \in R, \\ N(b) > N(r_0) > N(r_1) > \dots > N(r_n), \\ r_{n+1} = r_{n+2} = \dots = 0. \end{cases}$$

This $\{r_k\}$ is called **remainder sequence** for a and b .

Proposition 0.2.3. Let R be an Euclidean domain. Then, for any non-zero ideal $I \subseteq R$, $I = (d)$, where d is any non-zero element in I of minimum norm.

Proof. Let $I \subseteq R$ be a non-zero Ideal. Observe that the set $N[I \setminus \{0\}] = \{N(a) \mid a \in I, a \neq 0\}$ has a minimum element $m \in \mathbb{N} \cup \{0\}$ by the Well-ordering principle. Take $d \in N^{-1}(\{m\})$. Trivially, $(d) \subseteq I$. To show the reverse inclusion, let $a \in I$ be given. Since R possess a Division Algorithm, for some $q, r \in R$, $a = qd + r$ with $N(r) < N(d)$ or $r = 0$. But by the Minimality of $N(d)$, r must be zero. Therefore, $a = qd \in (d)$. $\square$

Theorem 0.2.4. Let R be an Euclidean domain, and a, b be non-zero elements in R . Let $\{r_k\}$ be a remainder sequence for a and b . Then, $(r_n) = (a, b)$.

Proof. Clearly, $r_n \mid r_{n-1}$ since $r_{n-1} = q_{n+1}r_n$. And $r_n \mid r_n$. If r_n divides r_{k+1} and r_k , then r_n divides $r_{k-1} = q_k r_k + r_{k+1}$. Inductively, r_n divides $b = r_{-1}$ and $a = r_{-2}$. This proved that $(a, b) \subseteq (r_n)$. To show the reverse inclusion, $a = q_0 b + r_0 \implies r_0 \in (a, b)$ and $b = q_1 r_0 + r_1 \implies r_1 \in (b, r_0) \subseteq (a, b)$. Inductively, $r_n \in (a, b)$. $\square$

0.3 Principal Ideal Domain (P.I.D.)

Definition 0.3.1. An Integral domain R is called **Principal Ideal Domain** if every Ideal in R is principal.

Directly, Euclidean domain $\implies$ P.I.D.

Proposition 0.3.2. Every nonzero prime ideal in a P.I.D. is a Maximal.

Proof. Let $(p) \subsetneq R$ be a Prime Ideal, $p \neq 0$. And, let $m \in R$ s.t. $(p) \subseteq (m) \subseteq R$. Enough to show that $(m) = (p)$ or $(m) = R$. By the assumption, $p = rm$ for some $r \in R$. And, since (p) is prime, $rm \in (p)$ implies $r \in (p)$ or $m \in (p)$.

- If $r \in (p)$, then $r = ps$ for some $s \in R$. $p = rm = psm$. Thus $sm = 1$, m is unit, $(m) = R$.
- If $m \in (p)$, then $(m) \subseteq (p)$, thus $(m) = (p)$.

□

0.4 Elements and Factorization

Definition 0.4.1. Let R be an Integral domain, let $r \in R$ be a nonzero, not a unit.

1. r is called **irreducible** if $r = ab \implies a$ or b is a unit.
2. p is called **prime** if $(p) \subseteq R$ is prime ideal. $(p \mid ab \implies p \mid a \text{ or } p \mid b)$.

Proposition 0.4.2. Prime element is irreducible in an integral domain.

Proposition 0.4.3. Irreducible element in P.I.D. is a prime.

In sum, in P.I.D., $\text{Maximal} \iff \text{Prime} \iff \text{Irreducible}$.

0.5 Unique Factorization Domain (U.F.D.)

Definition 0.5.1. A Unique Factorization Domain is an integral domain R in which every nonzero, nonunit $r \in R$ satisfies:

1. $r = p_1 p_2 \dots p_n$ (finite product of irreducibles).
2. Unique up to associates.

Theorem 0.5.2. Every Principal Ideal Domain is Unique Factorization Domain.

Proof. (Existence) Let R be a P.I.D. and $r \in R$. If r is irreducible, clear. Suppose r is reducible, $r = r_1 r_2$. If r_1, r_2 are not irreducible, continue. We get an ascending chain $(r) \subsetneq (r_1) \subsetneq (r_{11}) \subsetneq \dots \subseteq R$. In a P.I.D., every ascending chain is stationary (Noetherian property). Thus the process must terminate. $\square$

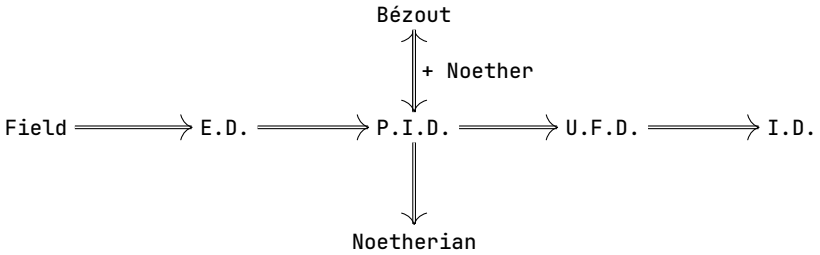
0.6 Bézout Domain

Definition 0.6.1. An integral domain R in which every ideal generated by two elements is principal is called a **Bézout Domain**. $(a, b) = (d)$.

Theorem 0.6.2. If R is Bézout domain and also U.F.D., then it is P.I.D.

Corollary 0.6.3. If R is Bézout domain and Noether domain, then R is P.I.D.

0.7 Summary Diagram (Page 12 Overview)



0.8 Detailed Examples

Example: $\mathbb{Z}[x]$ is U.F.D. but not P.I.D. The ideal $(2, x) = \{2p(x) + xq(x) \mid p, q \in \mathbb{Z}[x]\}$ is not principal. Suppose $(2, x) = (a(x))$. Then $a(x) \mid 2 \implies a(x) \in \{\pm 1, \pm 2\}$. If $a(x) = \pm 1$, then $(a(x)) = R$ (Contradiction, $(2, x)$ is proper since $1 \notin (2, x)$). If $a(x) = \pm 2$, then $a(x) \mid x$ which is impossible in $\mathbb{Z}[x]$. Thus $(2, x)$ is not principal.

Example: $\mathbb{Z}[\sqrt{-5}]$ is not U.F.D. $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. All factors $2, 3, 1 \pm \sqrt{-5}$ are irreducible but not associates. For instance, $N(2) = 4$. If $2 = \alpha\beta$, then $N(\alpha)N(\beta) = 4$. $N(a + b\sqrt{-5}) = a^2 + 5b^2$. $a^2 + 5b^2 = 2$ has no integer solution. Thus 2 is irreducible. However, $2 \mid (1 + \sqrt{-5})(1 - \sqrt{-5})$ but $2 \nmid 1 \pm \sqrt{-5}$. So 2 is irreducible but *not prime*.